DBS BANK: A TECH COMPANY GOING ALL IN ON AI

DBS Bank Limited (DBS), headquartered in Singapore, had attained many milestones towards its vision of becoming a technology driven, Artificial Intelligence (AI)-fuelled tech company by April 2023. More than a decade ago, the bank had embarked on a digital transformation journey, seeking to become digital to the core, by integrating technology into the customer experience, and adopting a start-up culture.[1] As part of this journey, DBS resolved to leverage AI-based machine learning to solve user problems and pain points. While the transition over the prior decade to becoming AI-driven had been successful, the company had realised that the endeavour towards embedding AI into all of its products and services was a continuing journey with moving goalposts.[2]

Before the transformation, DBS had a reputation of being 'Damn Bloody Slow' and not being customer friendly.[3] When Piyush Gupta (Gupta) joined the bank as its CEO in 2009, he realised that to stay relevant, the bank would have to think differently. He implemented an organisation wide drive for digital transformation and AI implementation, and garnered the company's board support to set aside a substantial budget for the transformation.[4] In the first few years of the transformation, DBS achieved a 9% compound annual growth rate (CAGR) in income and 13% CAGR in net profit.[5] This motivated the senior management to further invest in the transformation, and allowed the firm to diversify its business.[6]

Fast forward to 2023, the AI-led business transformation had resulted in several big wins for the bank, including a reputation for exceptional customer service and innovative digital solutions. DBS had reported a record performance as net profits grew 20% to US$6.02 (S$8.19) billion in the financial year 2022.[7] The company had invested US$0.74 (S$1) billion in technology annually over the past four years, with the fastest-growing investment pot allocated to enhancing data capabilities. The estimate was that the AI initiatives had generated US$110.25 (S$150) million in terms of additional revenue and US$18.38 (S$25) million from loss prevention and productivity gains in 2022 alone. Gupta believed that the economic value from AI fuelled growth could cross the billion mark in the next five years.[8]

Meanwhile, the banking industry had also evolved. The rise of digitalisation had intensified competition within the industry, prompting the Singapore government to introduce regulations enabling fully online banks to commence operations by 2022.[9] For example, Southeast Asian ride-hailing technology firm Grab had won a full digital bank licence and ventured into banking in 2022 in partnership with Singapore-based telecom provider SingTel.[10] Singapore had 98% banking penetration and most incumbents had strong digital product suites. Even so, analysts conjectured that technology companies would come into the market with aggressive product pricing and more innovative solutions and intensify the competition.[11] Gupta was however confident in his firm's transformation so far, and noted, "On the whole, I think we're relatively well positioned, and we should be able to hold our own".[12]

This case was written by Professor Emeritus Steven M. Miller and Lipika Bhattacharya at the Singapore Management University and Professor Thomas H. Davenport at Babson College. *The case was prepared solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.*

Copyright © 2023, Singapore Management University *Version: 2023-09-21*

DBS had used an 'All in on AI' approach to transform and infuse AI into every aspect of its business and invested significantly to enable the development and implementation of AI-based solutions.[13] DBS had also used AI to foray into new businesses such as a digital exchange and a global carbon exchange seeking new revenue streams.[14] However, future growth also demanded identifying and executing the right business opportunities.[15] But, what could these be?

Disruption, Traditional Banking, and its Decreasing Relevance

A key motivation for DBS in its journey towards becoming a tech company was the changing nature of traditional banking. This became more evident with the onslaught of the COVID-19 pandemic in 2020, when even traditional consumers quickly adapted to digital applications such as mobile banking and customer support through AI. A study across 30 countries noted that between February and June 2020, mobile banking usage had grown by 34%, while banking at branches had declined by 12%, leading analysts to predict that traditional banking would gradually disappear, and most banking services would get digitalised over the next five to seven years.[16]

The new wave of digital disruption had made banks vulnerable to losing market share to tech firms, which had already established themselves as financial services players.[17] Technology companies had started dabbling into financial services in the early 2000s and were able to compete on cost, forcing banks to lose margins as their services became increasingly commoditised.[18] By 2021, fintechs like PayPal, Alibaba, Squared, Grab, Adyen and Klarna had already achieved significant market caps, higher than some multinational banks.[19] For example, in November 2021, PayPal's market cap stood at US$267 billion, which at that time was nearly twice the market cap of Goldman Sachs, and larger than the market caps of all but a few of the very largest global financial institutions.[20]

Consumer behaviour in banking was also changing, as consumers were progressively able to compare mortgage rates, interest rates, and other banking deals online, picking the most favourable terms without regard to whether the product came from a traditional bank or a technology platform. As this approach to financial products gained traction, customers were able to enjoy much higher price transparency and direct comparability of products. A study conducted in 2021 on consumers in Asia Pacific had found that 62% customers preferred to begin a new banking relationship digitally, compared to 40% who were willing to visit a branch if required.[21]

As of August 2021, there were an estimated 250 digital banks worldwide, with 20% of them in the APAC region.[22,23] A key factor in favour of digital banking was the high cost of setting up and running physical bank branches, making physical-based service delivery less cost efficient. In a typical traditional retail bank, 30% of the total costs were incurred by its branch network while an additional 20% of its costs were incurred by its IT departments in maintaining legacy systems.[24] The average cost-to-income ratio of disruptor banks offering digital services was estimated to be 26% while that of traditional banks was estimated to be 60%.[25]

Most traditional banks had started to introduce standalone AI-based innovations such as chatbots for financial advisory, digital currency systems, variable pricing based on behavioural insights, facial recognition for ATM usage, and even loans based on an individual's earnings from peer-to-peer lending networks. However, these initiatives had been mostly incremental, independent changes that did not affect core processes in the banks. [26] Unlike these strategies, DBS had used a whole organisation approach to incorporate digitalisation and AI based solution building.

Digital Transformation at DBS

Originally known as the Development Bank of Singapore, DBS was established in Singapore as a public limited company to take over the industrial financing responsibilities of the country's Economic Development Board or EDB. It began operations in September 1968 and focused on financing Singapore's industrialisation and urban development projects. [27] Over the years, DBS greatly expanded its footprint throughout Asia through acquisitions and organic growth and became the largest bank in Southeast Asia.[28] However, as the company grew rapidly, its customer service lagged.

The Motivation

DBS's transformation journey was motivated by a few key factors. The company was ranked lowest in customer service amongst the largest banks in Singapore.[29] The bank was also facing institutional constraints in achieving acquisition-led growth and organic expansion in Asia, particularly in emerging Southeast and South Asia.[30] The bank was also encountering dwindling revenues and stagnant growth, and digital solutions presented new growth opportunities.[31]

However, at the time, DBS had 22,000 employees across different markets, and implementing organisational change was difficult.[32] Moreover, as most of the technology solutions were procured from external parties and technology talent was predominantly outsourced, it was cumbersome and slow to make technology-enabled changes to processes, products and services.[33]

Every Unit Approach

In late 2009, DBS embarked on a transformation journey and identified key areas that needed an overhaul including technology and culture. A whole organisation approach was used to execute the transformation strategy, which focused on three pillars — being digital to the core, embedding in the customer journey, and thinking and acting like a start-up company.[34] As the bank started embarking on its journey, AI emerged as a core capability and advantage that drove the transformation and helped the bank scale solutions quickly and effectively.[35]

While making the transition, DBS was careful in ensuring that its existing businesses were not neglected.[36] The mandate for digital transformation was applied across all organisational units by enforcing enthusiasm and vigour towards the new vision.[37] Even support functions like Audit and Human Resources (HR) were expected to digitalise their existing business and work processes, and these efforts brought about several benefits. For example, the Audit department reaped considerable productivity gains by first implementing data analytics, followed by machine learning-based techniques, to automate tasks such as branch risk profiling, trading fraud analysis, and credit risk assessment. Similarly, HR was able to improve employee satisfaction by using data-driven machine learning models to recognise key traits of top performers, anticipate potential attrition, and develop suitable intervention strategies.[38]

Overhauling Technology

DBS also reorganised its various business capability platforms to infuse business with technology. The first six years of the transformation journey were spent overhauling the technology and operations infrastructure to tighten IT alignment with the business. [39] During this phase, the technology and operations divisions were combined to form a single entity, with their reporting lines consolidated to improve operational excellence. Additionally, the bank's key processes were streamlined by applying lean principles. The core technology was revamped, and a new and improved

enterprise architecture was instated to support the diverse range of banking applications for different products, channels, functions, and integration layers.[40] A new core banking system was deployed across 12 markets in 28 months. Apps for Internet banking and mobile banking were launched to augment the bank's transaction banking capabilities. In addition, two separate councils (a Customer Experience Council to enhance customer engagement and an Innovation Council to foster an innovation culture) were set up with the intention of fostering a cultural mindset shift towards digitalisation, motivating experimentation and innovation within the organisation and introducing new digitally enabled customer experiences with the help of ideas from employees.[41]

Building an AI Advantage

The earliest efforts in AI in the bank included some failures that provided key learnings. For example in 2013, DBS collaborated with local research firm A*STAR on developing an AI lab under a three-year contract to explore AI applications with assigned data scientists from both firms.[42] The team worked on half a dozen projects, some of which were successful while others were not. Gupta described this portfolio of joint projects as "signalling tools" for the organisation. [43] DBS also conducted a large-scale pilot project with IBM's Watson for a wealth management application, but without a successful outcome.[44]

Through these setbacks, DBS learned that for successful project execution, agile approaches were a must and solutions had to be fine-tuned over time, with a minimal viable product as the starting point instead of delivering a fully functional elaborate solution after a significant lapse of time.[45]

New Framework

In 2015, after the core technology and operations infrastructure were in place, DBS began analysing key trends in emerging technologies and new competencies. A few key personnel from the bank visited the world's most renowned technology companies, including Google and Amazon, to learn about the industry's best practices. Around the same time, a catchy acronym was coined inside the bank to help galvanise the organisation around the transformation. Known as GANDALF (where the "D" signified DBS's ambition to be seen in the same league as tech titans Google, Amazon, Netflix, Apple, LinkedIn and Facebook)[46], the acronym was a rallying cry to motivate employees to think out of the box (refer to **Exhibit 1** for more information on GANDALF Cheese Wheel Framework).[47]

Creating Early Wins

DBS tried to create a few early wins, to drive support within the organisation towards the transformation (refer to **Exhibit 2** for details on some early win projects). In 2014, for example, DBS's ATM team worked with a data science team to create predictive models for preventive maintenance and cash replenishment at its ATMs. This effort helped reduce ATM downtimes, decrease out-of-cash occurrences by over 90%, the number of trips required to reload the machines by 10%, the amount of leftover cash returned to the bank by over 30% and enhance customer satisfaction.[48]

Another large-scale experiment towards digitalisation and AI solution building was a project called Digibank, launched first as an offering in the bank's India operations in 2016. DBS became the country's first mobile-only bank, as well as the first bank to make use of the national ID and biometric card (called Aadhaar) for all-digital on-boarding of new bank customers. [49] The launch and subsequent operation of Digibank helped transform the bank's approach to retail banking call centres and all aspects of retail customer service, not only in India, but in home turf Singapore, and in other

markets of operation as well.[50] The Customer Science team at Digibank India monitored the journey of every single Digibank customer on a real-time basis, using AI-based tools and capabilities. It proactively looked for indications of when the customer was struggling with using the mobile app, and developed intervention strategies to guide customers with the options on how to proceed with their journey. Putting these customer science capabilities in place enabled Digibank India to realise substantial productivity improvements, growth of its customer base by nearly six times, and growth in the number of financial transactions by over 12 times, while maintaining the same headcount at its customer contact centre over a four-year period.[51]

Hiring AI talent

DBS also tried to build its technology talent and infrastructure capabilities with the objective of switching from 85% outsourcing to 85% insourcing. The idea was to move away from outsourcing technology to building it in-house, and creating a 'tech DNA'. Starting in 2015, DBS went on a hiring spree, and began insourcing jobs that had previously been performed by third-party IT vendors. In 2017, for example, DBS tried to hire around 200 technology professionals including data scientists and AI experts.[52] The firm hired senior technologists with prior work experience in firms like Google, Yahoo and Huawei, to create cutting-edge solutions with application programming interfaces (APIs), cloud computing, site reliability engineering or SRE and expand ecosystems.[53] The talent focus was fondly referred to as 'building our timber' and comprised a three-pronged approach – to build tech DNA and capabilities; be the employer of choice for top tech talent; and upskill and reskill employees to be future-ready.[54]

Data First and Everything Goes to Cloud

AI-based solutions were highly dependent on appropriate and accessible data and DBS realised that to move forward with AI-enabled analytics, it needed to become a "Data First" organisation. The firm launched its Data First transformation programme in 2018 and defined it in terms of four pillars: 1) Analytics capability, 2) Culture and curriculum, 3) Enablement of data usage, and 4) Technology platforms. It viewed all four pillars as critical enablers for driving value from data, and also for providing the foundation to move beyond being a "Data First" company to an "All in on AI" company[55] (refer to **Exhibit 3** for a visual representation of Data First).

Towards this end, all legacy core banking applications were gradually moved onto modern platforms to ensure consistent, reliable, and scalable service delivery. In addition, most data servers were virtualised and a policy of "everything goes cloud" was instituted.[56] The policy of 'everything goes cloud' removed an over reliance on traditional data servers, enabling the bank to expand its computing power to handle ever-increasing amounts of data while simultaneously reducing its data centre footprint.[57]

Promoting Experimentation with AI

One of the strategies for AI to take root was to do a lot of experimentation and establish infrastructure and processes to improve the productivity and scalability of solutions. A key performance indicator (KPI) set was to conduct a thousand experiments a year, many of them involving data and AI. Events were held where these experiments were presented to encourage the bank's managers to think more deeply about how they could use the bank's data to deploy business-relevant AI solutions. Typically, around 20 or 30 most innovative projects were highlighted in these events. Gupta was personally involved with curating these events to showcase what he considered to be interesting experiments across the bank using AI as well as other technologies.[58]

Opening the Wallet

Gupta also "opened up the wallet" to AI experimentation by giving business units and functions the flexibility to hire quasi-data scientists within their own units (above and beyond the data scientists hired into the banks' Analytics Centre of Excellence) to see what they could accomplish. For example, using this approach, the HR division created a Job Intelligence Maestro (JIM) application, to help hire the most qualified people into the bank. The division also developed a personnel attrition prediction model, which used around 147 data inputs (like the time when employees came to work and the number of websites they visited) to derive the chances of attrition.[59]

Embedding AI in Customer Experience

Projects aimed at digitalising customer journeys were implemented rapidly. The focus was on moving away from branch banking to acquire customers through digital marketing, going paperless, removing customer pain points, enabling instant fulfilment through transaction automation, driving sticky customer behaviours and cross-selling through online engagement.[60]

DBS infused customer behaviour data with system data to build real-time analytical models to cater to digital customers' needs and enhance internal efficiencies. For example, the customer centre team was able to reduce inbound calls by over 14% in Singapore by analysing customers' transaction patterns, online behaviours and historical data to determine customers' receptiveness towards digital fulfilment. Similarly, the Institutional Banking Group was able to derive insights from supply-chain network data to deduce opportunities and risks in their customers' supply chain networks. These insights increased the bank's precision in acquiring customers, maximising value to customers and supplementing credit decisions with transaction data.[61] Gupta had noted,

> *This is very meaningful and material. And it's a far more intelligent and appropriate way to deal with customers: to give them choices, ideas, insights, and reflections. That's the big picture of where we are and what we are trying to do.*[62]

Automation Strategy

An enterprise-wide automation strategy and technologies such as Robotic Process Automation (RPA) were implemented to support and guide AI adoption in the organisation. In 2017, DBS partnered with IBM to form a Center of Excellence to provide end-to-end services for implementing and maintaining the RPA programme.[63] By 2020, the firm automated over 100 complex processes with RPA, which included tasks that were part of "intelligent automation" workflows employing machine-learning solutions for data-driven decision-making, optical character recognition, and voice recognition.[64]

While some data-driven projects were targeted towards building internal efficiencies, other projects were targeted towards improving customer experiences. AI-based solutions were implemented for customer focussed hyper-personalised marketing, sentiment analysis, credit decision-making, image and speech recognition, attrition and ATM cash outage predictions and other everyday banking activities. DBS also worked with regulators to increase the acceptability of these applications for decision-making activities.[65]

To improve the productivity of creating, deploying and maintaining AI models and the related process pipelines, the firm also created a new data platform called ADA, or "Advancing DBS with AI". The platform included a variety of capabilities for data ingestion, security, storage, governance, visualisation, and AI/analytics model management. The goal was to enable as much self-service as

possible within the bank for internal staff to create new AI models and maintain them over time. Many of the AI and analytics systems were migrated to hybrid clouds to enable faster deployment, scaling and processing.[66]

A separate team was formed to develop a new digital services platform that could foster partnership collaborations to offer new digital solutions – the idea was to build an ecosystem with partners to introduce various digital services for customers. The API platform comprised a repository of business, data, and infrastructural components to configure digital offerings. At the time of launch in 2017, the platform offered 150 APIs across more than 20 service categories such as fund transfers and real-time payments and had more than 50 participating companies on board including big household names like AIG (insurance), McDonald's (fast food), and Foodpanda (food delivery).[67]

New Business Avenues

DBS also used its AI focus to diversify its business and foray into new businesses. Separate teams were formed to actively innovate for new business growth opportunities and venture into non-bank businesses including e-marketplaces – like DBS Car Marketplace, DBS Electricity Marketplace, DBS Property Marketplace and Carousel. The DevOps pipeline was automated from testing to deployment, increasing the release cadence of new applications by tenfold.[68] DBS also entered into cross-referral agreements with regional peer-to-peer lending platforms like Funding Societies and Moolah Sense.[69]

A few experimentative projects were also launched to test new avenues. For example, in 2017, DBS implemented POSB Smart Buddy, the world's first in-school wearable tech savings and payments programme which involved creating an e-payments ecosystem within schools, to enable young students to cultivate sensible savings and spending habits. An accompanying smart buddy mobile app allowed parents to remotely manage their children's daily allowances, spending and savings, while empowering students to monitor their own finances.[70]

Reskilling Staff

Various training programs were implemented, using a combination of online and offline training, workshops, and mentorships, to equip employees with the necessary skills and resources to adapt to new ways of working or new forms of work. For example, customer centre employees who were no longer attending phone calls were reskilled to take on new job roles like live chat agents and customer experience designers. An environment of continuous learning was propagated as employees were trained to leverage data to improve the customer journey and business performance.

A curated curriculum called DigiFY was developed to impart seven digital skills – journey thinking, agile, data driven, digital business, digital technologies, digital communication, and risk and controls. These courses were offered to employees through a virtual university. By 2020, more than 80% of the staff had gone through the curriculum. Additionally, a six-month 'Data Heroes' programme was launched to build data analytics capabilities with an intensive, hands-on curriculum to equip employees with the skills to become data translators. Another program called 'hack2hire' was launched to bring in talent that would enhance DBS's digital offerings across a variety of skills, leveraging cloud technologies, machine learning, AI, big-data, and analytics; by mid-2020 'hack3hire' had engaged around 5,000 employees across different innovation programmes.[71]

Scaling AI Implementations

DBS used five key methodologies to intensify AI fuelling. Firstly, more than 200 data use cases were identified across the enterprise to deliver business value, with over 150 of these using advanced analytics or AI. Secondly, the advanced analytics and AI capabilities were reinforced through the dedicated efforts of more than 700 data and analytics experts engaged in these initiatives. Thirdly, a distinctive learning environment was established to foster a data-driven culture throughout the organisation. A curated e-learning curriculum and a comprehensive training programme were implemented to instil a data-driven mindset. Fourthly, ADA was used for accommodating big data storage and advanced analytics tools. Additionally, the technology focus tried to generate customer experiences that were relevant, reliable and safe. Moreover, governance processes and policies on responsible data usage were continually enhanced to capitalise on new business prospects and reinforce customer trust.[72]

The idea was to employ multiple value levers with individual use cases of AI (refer to **Exhibit 4** for All-in-on-AI value drivers).[73] While many other banks had opted to acquire AI capabilities from external vendors, Gupta had focused on building as many AI use cases as possible internally. By doing so, the bank aimed to have comparable capabilities to 'digital natives', i.e. technology-driven companies that were born in the digital age. Gupta's perspective was that by possessing similar AI capabilities to these digital natives, DBS Bank could effectively innovate and compete with them when necessary.[74]

By 2020, more than 90% of the organisation was using the ADA platform; 93% of existing applications were operating in the new environment and 99% of the applications had been migrated from physical servers to the virtual private cloud. DBS also moved much of its data from traditional data warehouses to data lakes[75] as the latter architecture was much cheaper to operate and better suited to less-structured data.[76] In addition, the bank created a new structure for its metadata, cleaned up 80 million incomplete data records, developed new protocols for who could access data and what customer data was suitable to capture, and introduced visualisation tools to make trends in data more apparent. The bank also created new governance structures for data, including a "Responsible Data Use Committee" that examined whether, or to what degree, customer-facing data was appropriate to collect and use in particular types of situations.[77]

By 2021, AI enablement was increasingly helping DBS strengthen its customer service, empowering employees to deliver their work more efficiently, and motivating more innovations within its ecosystem.[78] DBS Chief Analytics Officer, Sameer Gupta (Sameer), explained,

> *We created an in-house AI protocol that serves as a knowledge repository for all our AI use cases. It has more than 250 use cases, and anyone can search it. If someone is working on "attrition" for example, he/she can search for relevant tools, techniques, and data features. He/She doesn't have to build from scratch. Creating reusable assets was another focus. We wanted the low-code assets that the data teams have created to be leveraged in multiple use cases, including natural language processing.*[79]

Rewards

DBS saw revenue growth right from the early phases of its transformation journey (refer to **Exhibit 5** for DBS Revenue Growth over the years). The company's net income grew from US$4853 (S$6,603) million in 2009 to US$0.96 (S$1.3) billion by 2018 and US$1.65 (S$2.24) billion by 2022.[80]

The bank also ventured into non-bank businesses and invested in e-marketplaces, regional funding platforms and other payment platforms. In 2020, for example, new strategic partnerships with PayLah!

contributed to a three-fold uplift in transaction volumes, and DBS Property and Car Marketplaces generated more than US$0.74 (S$1) billion in loans across Singapore, Taiwan, and Hong Kong.[81] Despite these promising results, the revenues from these partnerships were not as per expectations, partly due to the negative impacts of the COVID-19 pandemic. The onwards plan was to double down efforts to sharpen focus on scaling these opportunities in the coming years (refer to **Exhibit 6** for other large companies using the 'All in on AI' approach).[82]

The transformation also enabled the firm to foray into new technologies seamlessly. In 2021, DBS launched a carbon credits platform called Climate Impact X (CIX) — a global carbon exchange marketplace to provide corporates with high-quality carbon credits for tackling climate change. Additionally, it contributed to the company's revenue stability during challenging times. Notably, in 2020, 2021 and 2022 DBS posted a total income of US$10,725 (S$14,592) million, US$10,585 (S$14,401) million, and US$12,129 (S$16,502) million respectively, despite several industries including banking, seeing tough financial years due to the pandemic.[83]

By 2023, DBS had created more than 100 AI and machine learning algorithms which analysed an internal data mart using 15,000 customer data points, to generate various types of nudges to improve customer service. For example, in Singapore alone, the bank was generating around 30 million hyper-personalised insights/nudges every month to engage its retail and wealth customers. Some examples of these AI-powered "nudges" included alerting customers about duplicated payments or higher-than-normal utility bills, and had led to a 10-12% uptick in customer responses.[84]

What Next?

While DBS had been an early mover towards being AI-driven, many other banks and companies across industries had also caught on and started to foray into similar transformations by 2021.[85] According to a Gartner survey, 69% of company board directors felt that the pandemic had accelerated their digital initiatives significantly. [86] For instance, Bank of America had reported receiving more deposits via mobile than its brick-and-mortar branches in 2020. [87] The size of transactions over mobile had also increased significantly, and many organisations had implemented sophisticated technologies like biometric identifications to securitise transactions. For example, ANZ bank had implemented Voice ID technology without additional authentication for customers to make payments of more than US$1,000 via mobile app (refer to **Exhibit 7** for more applications of AI in banking and finance).[88]

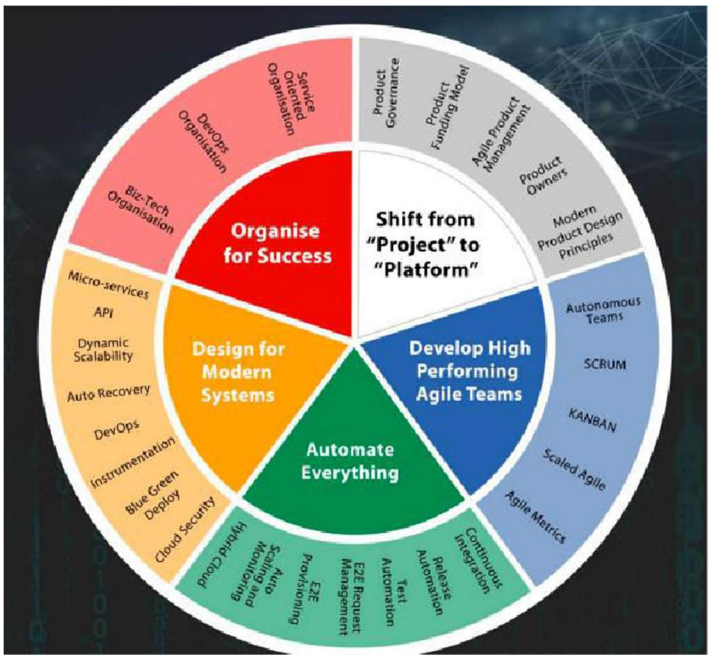
The All-in-on-AI approach had has enabled DBS to remain competitive amidst a changing banking landscape, by applying tech solutions to minimise latency in decision-making and deliver operational and business results. It had also helped the bank reduce costs by applying intelligent solutions to automate processes, tasks and interactions, and increase efficiency and predictive capabilities.[89] Moreover, it had helped the bank improve its customer service dramatically, and implement various new customer-targeted solutions.[90] For example, the bank introduced alerts to notify customers about favourable foreign currency rates to help customers invest better, location data to safeguard customers, or nudges to alert customers when they detected an odd card transaction.[91]

However, new risks continued to stem from emerging technologies like 5G, the Internet of Things (IoT), and advancements in AI - like Large Language Models (LLMs) or generative AI. These were considered to have the potential of changing customer experiences in the financial industry. Other emerging technologies like blockchain, distributed ledgers and digital currencies were expected to change the nature of the banking business (refer to **Exhibit 8** for more details on new and emerging technologies).

Moreover, customer experience was expected to become a pivotal competitive advantage and AI was anticipated to play a critical role in enabling predictions and personalised interactions for individual customers (refer to **Exhibit 9** for components of All-in-on-AI companies). But with the dramatic step changes in AI capabilities that had become available since early 2023, the path forward for how to use AI to gain competitive advantage and growth, higher productivity, and more innovation was unclear. With the burst of recent developments in AI, along with other big changes in the macro business and geo-political environments, it was looking a lot more like uncharted territory.

How could DBS compete with technology companies in a rapidly changing banking marketplace? How could it continue to operate like the 'digital natives'? How could it remain relevant in a competitive finance and banking industry? What opportunities could it bet on and how? How could AI help with those opportunities? Could the 'All in on AI' approach continue to provide the bank with a competitive advantage?

EXHIBIT 1: DBS GANDALF TRANSFORMATION CHEESE WHEEL



The DBS GANDALF acronym was formulated based on the firm's aspiration to building its own tech capabilities much like the big tech companies. G represented Google, A-Amazon, N-Netflix, A-Apple, L-LinkedIn, F-Facebook, while DBS aspired to be the D in GANDALF.

Source: Dave Gledhill, "Executing the digital strategy", DBS, SGX, Nov 2017, https://links.sgx.com/Executing_the_digital_strategy.ashx, accessed May 2023.

EXHIBIT 2: DBS EARLY WINS AI PROJECTS

IVR in Singapore

Retail customers in Singapore had the habit of making calls to the contact centre each time they had a question, even if that question could be conveniently addressed by the mobile app or the website.[92] This led to the design of the Interactive Voice Response (IVR) interception program, which allowed the bank to filter calls that could be directed to the mobile app. The retail customer contact centre catering to the local market in Singapore received around five to six million incoming interactions per year and over 80% of these interactions were phone calls, with the remainder comprising live chats and emails.[93] The IVR system was designed to query DBS's machine platform in real time about the customer and predict the reason for the call. A generic interception was put in place, allowing customers who fell under specific scenarios (e.g. credit card transaction was rejected) to be immediately routed to the bank's customer service officers, skipping through the tedious phone menu options. Within six months of operation, the redesigned menu shortened a customer's time spend on a call by 30 seconds, while customer satisfaction improved from 3.5 to 4.1 (with 5 being most satisfied). A few years into the implementation of the IVR program, call volumes reduced by 8%, indicating the project's success.[94] Additionally, to foster human-bot collaboration in the contact centre, a Quantum Image Recognition Application (QIRA) was implemented to seamlessly embed automation, with human experts brought in as and when needed.[95]

AML System

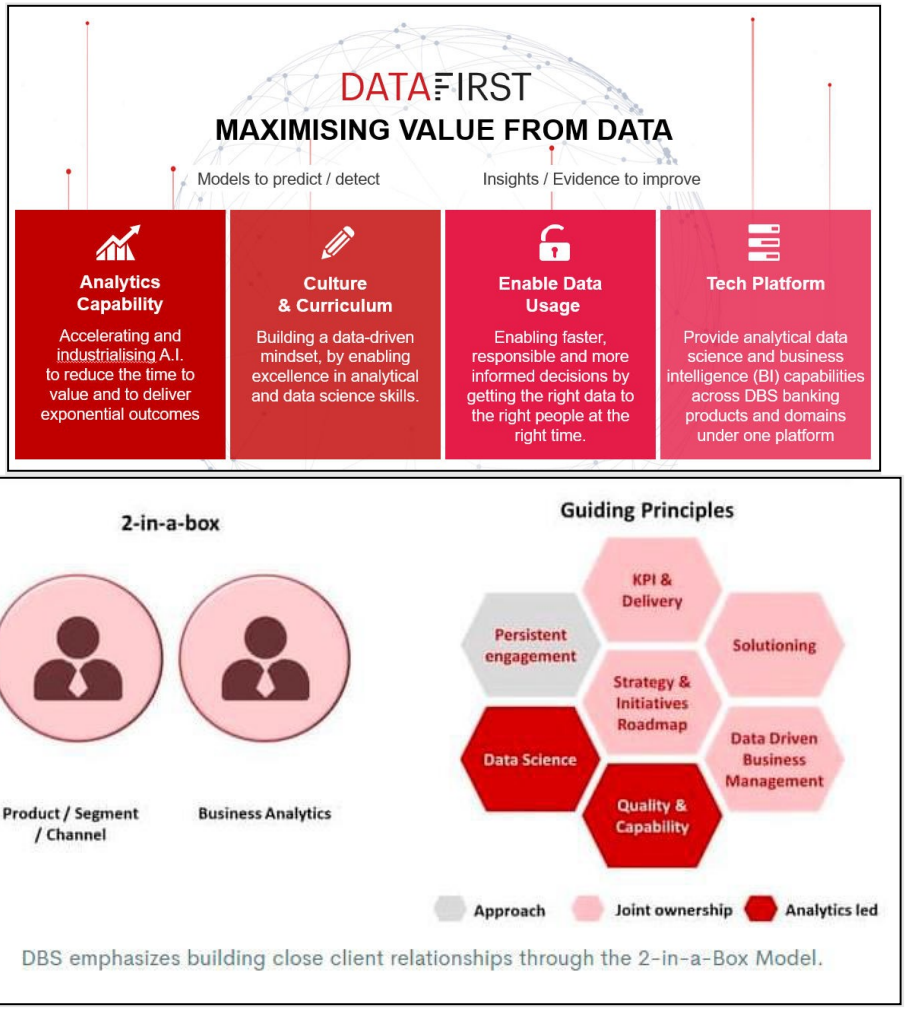
Another key project was implemented to use AI to prevent money laundering.[96] The existing rule-based systems were being used in the bank for managing anti-money laundering; however, one of their common shortcomings was that they created too many false positives (sometimes as many as 98%). The bank's transaction surveillance group created a machine-learning model that could filter alerts using the bank's data and prioritise the list of suspicious cases. Each case was given a "risk score," and the lowest risk cases were

put into "hibernation" for 90 days with the assumption that they did not need investigation, until their risk score increased. A "network link" was also developed using a graph database to analyse relationships among potential criminals, and a new system for the bank was built using open source technologies. Additionally, a case management system called 'Cruise' was implemented to replace the existing labour-intensive approach to investigation workflow. The Cruise platform integrated the outputs of the rule-based engine, the machine learning filter model, and the network link system, and provided the analysts with easy and integrated access to the relevant data from across the bank that was required for following up on the transactions they were investigating. While the rule-based systems could not be eliminated due to regulatory requirements, valued add systems like Cruise enabled analysts to combine the information from the machine learning system with the rule-based system to make better and faster decisions on money laundering detection. [97] While earlier, transaction surveillance analysts spent two or more hours looking into an alert, after implementing the above, they could do about a third more cases in the same amount of time, with higher accuracy.

Source: Author's own based on various published sources

EXHIBIT 3: DBS DATA FIRST FRAMEWORK

DBS tried to put data as a priority by making 'Data First', part of its company culture. To support the Data First culture, a two-in-a-box solution development structure (business and technology leads with joint KPIs) was created to motivate employees towards a common, holistic vision to deliver a differentiated customer experience.



DBS also conceptualised the PURE Framework to ensure responsible data use across the bank, where the bank assesses data use cases against its PURE Principles (Purposeful, Unsurprising, Respectful, and Explainable). The process was a key enabler in ensuring that data use was responsible and that data privacy considerations were embedded within each unit's culture and business practices.

- **P stands for Purposeful** - Data use should be Purposeful
- **U stands for Unsurprising**- Data use should be expected by individuals and/or corporates
- **R stands for Respectful** - Data use should be respectful, considering social norms
- **E stands for Explainable** – Data use should be explainable as to how and why we are doing it

Source: DBS Annual Report 2020

EXHIBIT 4: TYPICAL VALUE DRIVERS OF 'ALL IN ON AI' COMPANIES

Speed to execution: Apply AI to accelerate time to operational and business results by minimising latency in decision-making and action.

Cost reduction: Apply AI to intelligently automate business processes, tasks, and interactions to reduce cost, increase efficiency, improve environmental sustainability, and ensure predictability.

Comprehension of complexity: Apply AI to improve understanding and decision-making by deciphering patterns, connecting the dots, and predicting outcomes from increasingly complex data sources.

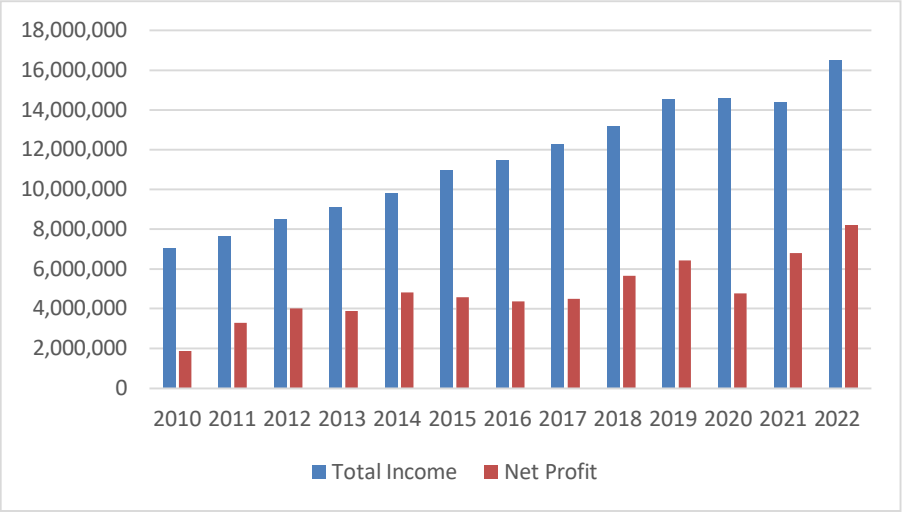
Transformed engagement: Apply AI to change how customers and employees interact with smart systems to expand means of engagement via voice, vision, text, and touch.

Fuelled innovation: Apply AI to generate deep insights into where to play and how to win, enabling the creation of new products, market opportunities, and business models.

Fortified trust: Apply AI to secure one's brand from risks such as fraud, waste, abuse, and cyber intrusion, consequently assuring stakeholders and enhancing trust amongst customers.

Source: Thomas H. Davenport and Nitin Mittal, "All in on AI", Harvard Business Review Press, 2023. pg. 23.

EXHIBIT 5: DBS REVENUE GROWTH



Source: Annual Reports

EXHIBIT 6: EXAMPLES OF TECH COMPANIES THAT ARE 'ALL IN ON AI'

Company	'All in on AI' approach
Loblaw	Loblaw, Canada's largest grocery chain, expanded to become the largest software platform for healthcare in the country. Its AI-enabled health app helps Canadians to navigate effectively through the healthcare system and provides a "front door" to health services. It also uses AI capabilities to provide nutrition information and healthy product recommendations for the 55,000 products sold in its retail stores.
Anthem	Anthem, a large health insurance company in the US, uses an AI enabled platform to connect its members digitally with the healthcare providers and services they need.
Calico	Calico leverages AI and machine learning to analyse large amounts of data in its research to identify new targets for intervention in aging-related diseases.
CCC Intelligent Solutions	CCC is a midsize company that dominates the supply of data and AI-based image analysis for automobile insurance collision damage assessment.
Toyota	Toyota has been focusing on using AI to make human driving smarter and safer rather than making fully autonomous vehicles in order to reduce auto accident fatalities, especially for teens and others driving recklessly. In addition, its AI-enabled advanced driver assistance system offers semi-autonomous cruise and parking capabilities.
Airbus	Airbus leverages AI technologies across various areas of its business to improve efficiency, safety, and customer experience in the aviation industry.
Shell	Shell has implemented AI for exploration and production, refining and chemicals, supply chain, safety and sustainability.
Deloitte	Deloitte uses AI across audit and assurance, tax and legal services, consulting, risk management, and operations.
Ping An	A Chinese insurance company that used AI to create new business models, strategies, ecosystems, and processes. The Good Doctor platform creates a new business in healthcare; the company developed several AI platforms to create new businesses from new use case scenarios.
Progressive	The car insurance company uses machine learning models to translate driving behaviours into the prices it charges individual customers; also uses AI analysis of mobile phone images to speed up the claims processing— and to help other aspects of the insurance process
Unilever	Unilever uses AI across the breadth of the organisation, in product development, marketing, supply chain management, quality control, and customer service.

Source: Thomas H. Davenport and Nitin Mittal, "All in on AI", Harvard Business Review Press, 2023.

EXHIBIT 7: EXAMPLES OF APPLICATIONS OF AI IN BANKING AND FINANCE

Areas	Application of AI
Fraud detection	AI algorithms are used to detect and prevent fraudulent transactions.
Risk assessment	AI is used to analyse large amounts of data to assess credit and loan risks.
Algorithmic trading	AI algorithms are used for high-frequency trading, risk management, and personalising portfolio.
Credit scoring	AI is used to analyse vast amounts of data to assess creditworthiness and make loan decisions.
Portfolio management	AI-powered investment advice and portfolio management services.
Operations	AI helps to automate repetitive tasks and improve efficiency in areas such as back-office processing and compliance.
Financial planning	AI-powered financial advisors provide personalised investment advice.
Marketing and sales	AI-driven personalised recommendations help banks cross-sell products.
Customer experience and service	AI-powered chatbots provide 24/7 customer support and handle basic queries.

Source: DBS Bank, "Artificial intelligence: the future of banking & finance", https://www.dbs.com/artificial-intelligence-machine-learning/artificial-intelligence/ai-the-future-of-banking-and-finance.html, accessed 26 May 2023.

EXHIBIT 8: NEW AND EMERGING TECHNOLOGIES IN THE FINANCIAL ARENA

There were several new and emerging technologies that were perceived to be influential in changing the financial industry in the near future. Some of the most relevant of these technologies included AI advancements, 5G, IoT, blockchain, and generative AI.

5G offered greater transmission speeds and lower latency, making it easier for networks to support connected devices, bandwidth-intensive apps, remote operations and virtual networks.[98] IoT on the other hand had started to become prevalent with the rise of smart devices and machine-to-machine connectivity without human intervention. This could mark the beginning of machine-based sensor communications in smart cities and widespread democratisation of the use of data.[99]

Separately, there was a dramatic step change in AI capabilities demonstrated with Open AI's release of GPT-3 in 2020, with that capability substantially amplified by the company's subsequent release of ChatGPT towards the end of 2022 and GPT-4 in early 2023[100]. The GPT series of AI models and versions of ChatGPT based on these newer GPT-X models, along with similar models from other organisations, were all based on Large Language Models (LLMs), also called generative or foundation models. This same underlying approach of using these models pre-trained on vast amounts of unstructured data (text or images, video, audio, computer code, or any combination thereof)[101] and that use a relatively recent AI model and training approach based on the transformer architecture, has resulted in a proliferation – a "Cambrian Explosion" magnitude of expansion – of generative AI capabilities and applications that are being quickly and widely deployed across every sector of the economy, especially banking and finance.[102]

Source: Authors' own

EXHIBIT 9: COMPONENTS OF "AI-FUELED" COMPANIES THAT ARE "ALL IN ON AI"

- Broad enterprise adoption of AI, using multiple technologies
- Many AI systems in production deployment
- Employing AI to reimagine and reengineer work processes
- A high percentage of the organisation fluent in AI and how it can be applied
- Long term commitments to and investment in AI
- Unique and voluminous sources of data, analysed and acted upon in real time
- A framework for ethical and trustworthy AI in place
- The organisation becomes a learning machine

Source: Thomas H. Davenport and Nitin Mittal, "All in on AI", Harvard Business Review Press, 2023, pp 7-125.

References

1 Annie Koh, Robin Speculand, Adina Wong and Wee-Kiat Lim, "DBS: Digital Transformation to Best Bank in the World", Singapore Management University, July 2020, https://cmp.smu.edu.sg/case/4426, accessed April 2023.
2 DBS, Technology of the Future, "Redefining the Future of Banking as a Technology Company", November 1, 2022, https://www.dbs.com/technology-future/dbs-redefining-the-future-of-banking-as-a-technology-company.html, accessed April 2023.
3 Thomas H. Davenport and Randy Bean, "Portrait of an AI Leader: Piyush Gupta of DBS Bank", MIT Sloan Review, August 31, 2021, https://sloanreview.mit.edu/article/portrait-of-an-ai-leader-piyush-gupta-of-dbs-bank/, accessed April 2023.
4 Ibid.
5 Sia Siew Kien, Peter Weill, and Mou Xu, "DBS: From the 'World's Best Bank' to Building the Future-Ready Enterprise", https://store.hbr.org/product/NTU231, accessed March 2022.
6 Ibid.
7 Reuters, "Singapore bank DBS Q4 profit jumps 69%, says rate rises to moderate", CNA, Business, February 13, 2023, https://www.channelnewsasia.com/business/singapore-bank-dbs-q4-profit-jumps-rate-rises-moderate-3273506, accessed April 2023.
8 Kelly Ng, "Piyush Gupta Counting on DBS' AI and Data Capabilities to Pull Ahead of Peers", DBS, October 31, 2022, https://www.dbs.com/artificial-intelligence-machine-learning/artificial-intelligence/, accessed April 2023.
9 Monetary Authority of Singapore, "MAS Announces Successful Applicants of Licenses to Operate New Digital Banks in Singapore", December 4, 2020, https://www.mas.gov.sg/news/media-releases, accessed April 2023.
10 Anshuman Daga, "DBS CEO Says Tough for Digital Banks to Muscle into Singapore", Reuters, December 3, 2021, https://www.reuters.com/technology/dbs-ceo-says-hard-digital-banks-gain-market-share-singapore-2021-12-03/, accessed January 2022.
11 Sheila Chiang, "Singapore's Digital Banks Dangle Incentives to Win New Customers — is it Sustainable?", CNBC, February 20 2023, https://www.cnbc.com/singapores-digital-banks-dangle-incentives-to-win-new-customers-is-it-sustainable.html, accessed April 2023.
12 Ibid.
13 Thomas H. Davenport and Nitin Mittal, "All in on AI: How Smart Companies Win Big with Artificial Intelligence", Harvard Business Review Press, January 24, 2023, https://www.amazon.com/All-AI-Companies-Artificial-Intelligence-ebook/dp/B09Q6KHWVT, accessed January 2023.
14 Anshuman Daga, "DBS CEO Says Tough for Digital Banks to Muscle into Singapore", Reuters, December 3, 2021, https://www.euronews.com/next/2021/12/03/reuters-next-dbs, accessed April 2023.
15 DBS, "Artificial Intelligence: The Future of Banking & Finance", April 18, 2023, https://www.dbs.com/artificial-intelligence-machine-learning/artificial-intelligence/ai-the-future-of-banking-and-finance.html, accessed April 2023.
16 Jörg Erlebach, Marc Pauly, Lucas Du Croo De Jongh and Michael Strauß, "The Sun is Setting on Traditional Banking", BCG, November 24, 2020, https://www.bcg.com/publications/2020/bionic-banking-may-be-the-future-of-banking, accessed January 2022.
17 Mariano Belinky, Emmet Rennick and Andrew Veitch, "The Fintech 2.0 Paper: Rebooting Financial Services", Oliver Wyman, June 2015 https://www.oliverwyman.com/The_Fintech_2_0_Paper.pdf, accessed March 2022.
18 Ibid.
19 Naeem Aslam, "Paypal's Market Cap is Almost Double Goldman Sachs; Why Wall Street Needs to Pay Attention", NASDAQ, November 8, 2021, https://www.nasdaq.com/articles/paypals-market-cap-is-almost-double-goldman-sachs-why-wall-street-needs-to-pay-attention, accessed June 2023.
20 Ibid.
21 Fintech News Singapore, "In APAC, Southeast Asia Consumers are Most Receptive to Challenger Banks", April 1, 2021, https://fintechnews.sg/49973/virtual-banking/in-apac-southeast-asia-consumers-are-most-receptive-to-challenger-banks/, accessed January 2022.
22 Jungkiu Choi, Prasanna Santhanam, Pauline Wray, Shobhit Shubhankar and Jeroen Vandensteen, "The Rise of Digital Banking in Southeast Asia", BCG, https://media-publications.bcg.com/The-Rise-of-Digital-Banking-in-Southeast-Asia.pdf, accessed January 2023.
23 Ibid.
24 Ibid.
25 Colm Farrell, "Monzo: A Digital Offering that has Traditional UK Banks Scrambling", Harvard Business School Digital Initiative, February 9, 2020, https://d3.harvard.edu/platform-digit/submission/monzo-a-digital-offering-that-has-traditional-uk-banks-scrambling/, accessed January 2023.
26 Ibid.
27 Singapore Infopedia, "Development Bank of Singapore", December 1, 2015, https://eresources.nlb.gov.sg/infopedia/articles/SIP_2015-12-01_132040.html , accessed January 2023.

28 Thomas H. Davenport and Randy Bean, "Portrait of an AI Leader: Piyush Gupta of DBS Bank", MIT Sloan Review, August 31, 2021, https://sloanreview.mit.edu/article/portrait-of-an-ai-leader-piyush-gupta-of-dbs-bank/, accessed April 2023.
29 Ibid.
30 Ibid.
31 Annie Koh, Robin Speculand, Adina Wong and Wee-Kiat Lim, "DBS: Digital Transformation to Best Bank in the World", Singapore Management University, July 1, 2020, https://cmp.smu.edu.sg/case/4426, accessed April 2023.
32 Ibid.
33 Ibid.
34 Annie Koh, Robin Speculand, Adina Wong and Wee-Kiat Lim, "DBS: Digital Transformation to Best Bank in the World", Singapore Management University, July 1, 2020, https://cmp.smu.edu.sg/case/4426, accessed April 2023.
35 Douglas Blakey, "Mastering the AI Advantage: How DBS Transformed into a Digital Leader", Retail Banker International, January 30, 2023, https://www.retailbankerinternational.com/analysis/how-dbs-transformed-into-a-world-class-digital-leader/?cf-view&cf-closed, accessed April 2023.
36 Siew Kien Sia, Peter Weill and Nila Zhang, "Designing a Future-Ready Enterprise: The Digital Transformation of DBS Bank", California Management Review, 63(3), March 5, 2021, https://journals.sagepub.com/doi/, accessed April 2023.
37 Ibid.
38 Ibid.
39 Ibid.
40 Sia Siew Kien, Christina Soh, Peter Weill and Yvonne Chong, "Rewiring the Enterprise for Digital Innovation: The Case of DBS Bank", Nanyang Business School, June 2, 2015, https://Case%20Study.pdf, accessed August 2023.
41 Ibid.
42 Xiaoli Li, "Data Analytics – from Research to Real-World Impact", A*Star Institute of Infocomm Research (I2R), May 18, 2021, https://ims.nus.edu.sg/wp-content/uploads/2021/06/Xiaoli-Li.pdf, accessed April 2023.
43 Thomas H. Davenport and Randy Bean, "Portrait of an AI Leader: Piyush Gupta of DBS Bank", MIT Sloan Review, August 31, 2021, https://sloanreview.mit.edu/article/portrait-of-an-ai-leader-piyush-gupta-of-dbs-bank/, accessed April 2023.
44 Eileen Yu, "DBS Bank is IBM's first Watson Customer in Asia", ZDNet, January 9, 2014, https://www.zdnet.com/finance/dbs-bank-is-ibms-first-watson-customer-in-asia/, accessed April 2023.
45 Robin Speculand, "A Three Step Model for Building Ecosystems: Lessons from the World's Best Bank", DBS, Business Ecosystem Alliance, https://business-ecosystem-alliance.org/2021/12/06/a-three-step-model-for-building-ecosystems-lessons/, accessed April 2023.
46 Tuber, "Why DBS wants to be Gandalf", Public Service Division - Challenge, April 5, 2018, https://www.psd.gov.sg/challenge/ideas/feature/why-dbs-wants-to-be-gandalf#: accessed June 2023.
47 Dave Gledhill, "Executing the Digital Strategy", DBS, November 17, 2017, https://Executing_the_digital_strategy, accessed April 2023.
48 Finextra, "DBS Recognised for Innovations in ATM Technology", November 28, 2014, https://www.finextra.com/pressarticle/57870/dbs-recognised-for-innovations-in-atm-technology, accessed April 2023.
49 Thomas H. Davenport and Steven M. Miller, "Working with AI: Real Stories of Human-Machine Collaboration", The MIT Press, September 27, 2022, https://mitpress.mit.edu/9780262047241/working-with-ai/, accessed April 2023.
50 Tom Davenport "The Future of Work Now: Customer Science at DBS Bank", Forbes, April 7, 2021, https://www.forbes.com/sites/tomdavenport/2021/04/07/the-future-of-work-now-customer-science-at-dbs-bank/?sh=21cc7b5f3edc, accessed April 2023.
51 Ibid.
52 Simon Mortlock, "DBS MD: 'I'm hiring 200 techies in 12 months'", efinancialcareers, May 22, 2017, https://www.efinancialcareers.sg/news/2017/05/dbs-technology-jobs-soh-siew-choo, accessed April 2023.
53 DBS Group, "Transforming DBS Bank into a Tech Company", Banking Innovation. December 3, 2020, articles/transforming-dbs-bank-tech-company, accessed April 2023.
54 DBS, Annual Report 2020.
55 Sameer Gupta, "A Look at How DBS is Becoming a Data-Driven Bank," presentation to the QuantumBlack AIxImpact Conference, Singapore, February 18, 2022, https://www.youtube.com/watch?v=gPxHNup_zCk, accessed June 2023.
56 Sia Siew Kien, Peter Weill and Mou Xu, "DBS: From the 'World's Best Bank' to Building the Future-Ready Enterprise", MIT Center for Information Research, March 18, 2019, https://store.hbr.org/product/NTU231, accessed April 2023.
57 Ibid.
58 Tom Davenport, "The Future of Work Now: Customer Science at DBS Bank", Forbes, April 7, 2021, https://www.forbes.com/sites/tomdavenport/2021/04/07/the-future-of-work-now-customer-science-at-dbs-bank/?sh=21cc7b5f3edc, accessed April 2023.

59 Ibid.
60 Annie Koh, Robin Speculand, Adina Wong and Wee-Kiat Lim, "DBS: Digital Transformation to Best Bank in the World", Singapore Management University, July 1, 2020, https://cmp.smu.edu.sg/case/4426, accessed April 2023.
61 DBS, Annual Report 2018.
62 Kelly Ng, "Piyush Gupta Counting on DBS' AI and Data Capabilities to Pull Ahead of Peers", DBS, October 31, 2022, https://www.dbs.com/artificial-intelligence-machine-learning/artificial-intelligence/, accessed April 2023.

63 DBS, "DBS Bank Accelerates Digitalisation Transformation with Robotics Programme", November 30, 2017, DBS_Bank_accelerates_digitalisation_transformation_with_robotics_programme, accessed April 2023.
64 Anand Singh Chandel, "How to Make Banking 'Invisible' with Digital Transformation", IBM, April 9, 2019, https://www.ibm.com/blogs/client-voices/banking-invisible-digital-transformation/, accessed April 2023.
65 Thomas H. Davenport and Randy Bean, "Portrait of an AI Leader: Piyush Gupta of DBS Bank", MIT Sloan Review, August 31, 2021, https://sloanreview.mit.edu/article/portrait-of-an-ai-leader-piyush-gupta-of-dbs-bank/, accessed April 2023.
66 Achim Granzen, "DBS Bank Followed Four Design Principles When Building its Enterprise Data Platform," Forrester, July 5, 2020, https://www.forrester.com/blogs/dbs-bank-followed-four-design-principles-when-building-their-enterprise-data-platform/, accessed April 2023.
67 Ibid.
68 DevOps is a globally known set of practices that combines software development and IT operations. It aims to shorten the systems development life cycle and provide continuous delivery with high software quality.
69 Robin Speculand, "A Three Step Model for Building Ecosystems: Lessons from the World's Best Bank", DBS, Business Ecosystem Alliance, https://business-ecosystem-alliance.org/2021/12/06/a-three-step-model-for-building-ecosystems-lessons/, accessed April 2023.
68 DBS, "POSB Smart Buddy - World's First In-School Wearable Tech Savings & Payments Programme Launches in 19 Schools", August 16, 2017, Worlds_first_in_school_wearable_tech_savings_and_payments_programme, accessed April 2023.
71 Banking Innovation, "DBS Bank: An All-Encompassing Innovation Leader", August 27, 2020, https://bankinginnovation.qorusglobal.com/content/reports/dbs-bank-all-encompassing-innovation-leader, accessed January 2023.
72 DBS, Annual Report 2021.
73 Thomas H. Davenport and Randy Bean, "Portrait of an AI Leader: Piyush Gupta of DBS Bank", MIT Sloan Review, August 31, 2021, https://sloanreview.mit.edu/article/portrait-of-an-ai-leader-piyush-gupta-of-dbs-bank/, accessed April 2023.
74 QuantumBlack AI by McKinsey, "Power Up: How Southeast Asia's Largest Bank is Becoming AI-Fueled", November 29, 2022, https://www.mckinsey.com/power-up-how-southeast-asias-largest-bank-is-becoming-ai-fueled, accessed April 2023.
75 A data lake is a centralised repository designed to store, process, and secure large amounts of structured, semi-structured, and unstructured data.
76 Ibid.
77 ET Edge Insights, "Reimagining Banking through Transformative Digitization", June 16, 2020, https://etinsights.et-edge.com/reimagining-banking-through-transformative-digitization/, accessed April 2023.
78 DBS Annual Report 2021.
79 QuantumBlack AI by McKinsey, "Power Up: How Southeast Asia's Largest Bank is Becoming AI-Fueled", November 29, 2022, https://www.mckinsey.com/power-up-how-southeast-asias-largest-bank-is-becoming-ai-fueled, accessed April 2023.
80 DBS, Annual Report 2022.
81 DBS, Annual Report 2020.
82 Ibid.
83 DBS, Annual Report 2020, Annual Report 2021 and Annual Report 2022.
84 Kelly Ng, "Piyush Gupta Counting on DBS' AI and Data Capabilities to Pull Ahead of Peers", DBS, October 31, 2022, https://www.dbs.com/artificial-intelligence-machine-learning/artificial-intelligence/, accessed April 2023.
85 Vladimir Lugovsky, "Digital Transformation in Banking: How to Make the Change", Forbes, December 29, 2021, https://www.forbes.com/sites/forbesbusinesscouncil/2021/12/29/, accessed January 2023.
86 Laurence Goasduff, "COVID-19 Accelerates Digital Strategy Initiatives", Gartner, November 6, 2020, https://www.gartner.com/smarterwithgartner/covid-19-accelerates-digital-strategy-initiatives, accessed April 2023.
87 Deloitte Insights, "Accelerating Digital Transformation in Banking", 2018, https://www2.deloitte.com/content/dam/Deloitte/jp/Documents/financial-services/bk/en-bk-global-banking-survey.pdf, accessed January 2023.
88 Financial Review, "ANZ's Drawn-Out Tech Transformation", December 8, 2021, https://www.afr.com/chanticleer/anz-s-drawn-out-tech-transformation-20211207-p59feg, accessed January 2022.

89 Tom Groenfeldt, "DBS Leads in using AI and ML While Maintaining Human Touch", Forbes, December 6, 2022, https://www.forbes.com/sites/tomgroenfeldt/2022/12/06/dbs-leads-in-using-ai-and-ml-while-maintaining-human-touch/?sh=6f3c54d4354b, accessed April 2023.
90 Ibid.
91 Ibid.
92 Ibid.
93 Ibid.
94 Tom Davenport, "The Future of Work Now: Customer Science at DBS Bank", Forbes, April 7, 2021, accessed April 2023.
95 Thomas H. Davenport and Steven M. Miller, "Working with AI: Real Stories of Human-Machine Collaboration", The MIT Press, September 27, 2022, https://mitpress.mit.edu/9780262047241/working-with-ai/, accessed April 2023.
96 Tom Davenport, "The Future of Work Now: AI-Driven Transaction Surveillance at DBS Bank", Forbes, October 23, 2020, https://www.forbes.com/sites/tomdavenport/2020/10/23/the-future-of-work-now-ai-driven-transaction-surveillance-at-dbs-bank/?sh=1d019cc13f7f , accessed April 2023.
97 Ibid.
98 DBS, Annual Report 2020.
99 Ibid.
100 OpenAI, "Product Overview", https://openai.com/product, accessed 27 May 2023.
101 Jeanina Casusi, "What is a Foundation Model? An Explainer for Non-Experts", Institute for Human-Centered Artificial Intelligence (HAI), Stanford University, May 10, 2023, https://hai.stanford.edu/news/what-foundation-model-explainer-non-experts, accessed 26 May 2023.
102 Cem Dilmegani, "10+ Generative AI Finance Use Cases in 2023," AIMultiple, July 26, 2023, https://research.aimultiple.com/generative-ai-finance/, accessed 27 May 2023.